

E-Waste Triage Tool + Contamination Validation

Non-Destructive Value/Hazard Prediction, Validated Against Real Leaching Behavior — Full Project Plan

One-Sentence Summary

You're building a cheap gadget-and-code tool that guesses how valuable and how dangerous an old electronic device is without opening it, then proving your "dangerous" guesses are actually true by soaking those devices in water and testing whether they really leak measurable contamination — like real rain would on a landfill.

Research Question

Can a low-cost, non-destructive sensing pipeline predict a device's recoverable material value and hazard risk — and does that hazard classification actually correspond to measurable real-world contamination when devices are exposed to simulated environmental conditions (rain/landfill leaching)?

Why this matters (your framing for judges): Current e-waste triage requires either destructive teardown or expensive lab assay. A validated non-destructive tool would let communities sort e-waste cheaply — but a hazard "label" is only meaningful if it predicts something that actually happens in the environment. This project builds the engineering tool, then scientifically validates it against real leaching/contamination behavior, rather than treating the label as the finish line.

Project Structure (Two Parts)

Part 1 — Engineering: a sensor + ML pipeline that non-destructively predicts device value and hazard category.

Part 2 — Environmental Science: a leaching experiment testing whether "hazardous" devices actually release measurable contamination into water, and whether "safe" devices don't — validating that your engineering classification reflects real environmental risk, not just a guess.

Steps To Take Soon (This Week)

- 1. Order your sensor components** - Digital multimeter with continuity/resistance mode (~\$15-20) - Small handheld magnet or Hall-effect sensor module (~\$5-10) - Precision scale, 0.01-1 g resolution (~\$15-20) - USB webcam or phone camera for the imaging component (likely already owned)
 - 2. Order water contamination test supplies** - Heavy metal test strips (lead/copper detection — sold for aquarium/home water testing, ~\$15-25 for a multi-test kit) - pH test strips (~\$5) - Clear containers/jars for soaking devices (~\$10)
 - 3. Start collecting your validation device sample** Gather 15-25 small electronics you already have or can easily get: chargers, mice, keyboards, cables, remotes, small speakers, old phones if available.
 - 4. Reach out to a real-world validation partner — start this now, longest lead time** Email or call your school, a local library, community center, or e-waste collection event/recycler and ask to run a non-destructive scan on a batch of their incoming e-waste for your science fair project.
 - 5. Install Python + basic ML libraries** scikit-learn for structured-data models; add PyTorch/torchvision if including image-based photo classification as an input.
-

Phase 1 — Build Your Non-Destructive Sensing Protocol (Weeks 1-2)

1. For each device, record:
 - Weight (grams)
 - Magnetic response
 - Basic conductivity/resistance at standard probe points
 - A standardized photo (consistent lighting/background)
 - Observable metadata (device category, approximate age)
2. Log everything in a spreadsheet with a unique ID per device.

Checkpoint: A structured dataset of non-destructive readings for every device, before opening any of them.

Phase 2 — Ground Truth via Manual Teardown (Weeks 2-4)

1. Using only a screwdriver (no cutting tools, no chemicals), open each device.
2. Record what's inside: battery presence/type, circuit board size/complexity, visible hazardous components (leaded solder, button-cell batteries, mercury switches), and weight of separated recoverable-metal components vs. plastic housing.
3. Assign each device a **ground truth value score** (using published average scrap prices for copper/gold-bearing boards) and a **ground truth hazard flag** (yes/no plus category).

Checkpoint: Every device has a paired non-destructive reading set and verified ground truth.

Phase 3 — Build & Validate the Prediction Model (Weeks 4-6)

1. Use cross-validation (e.g., leave-one-out) given your likely small sample size — legitimate and expected for a dataset this size, and worth stating explicitly in your writeup.
2. Train a regression model (value) and classification model (hazard flag) using your non-destructive readings.
3. Start with a random forest or gradient boosting model — handles structured sensor data well and is interpretable.
4. Report R^2 /mean absolute error (value) and precision/recall (hazard classification).
5. Identify which readings matter most (feature importance) — a genuinely useful standalone finding.

Checkpoint: A trained, validated model with documented accuracy metrics.

Phase 4 — Environmental Science Layer: Leaching Validation (Weeks 6-8)

This is the new science-validation phase that turns your hazard label into a tested environmental claim.

1. Select a subset of your teardown devices spanning both hazard categories: some flagged “hazardous” (leaded solder, batteries present) and some flagged “safe.”
2. For each selected device (or its opened components, to maximize water contact with exposed solder/metal), submerge in a fixed volume of distilled water in a labeled clear container.
3. Simulate a realistic landfill/rain exposure window — soak for a consistent period (e.g., 48-72 hours), keeping temperature and container conditions the same across all samples.
4. After soaking, test each container’s water using your heavy metal test strips (lead/copper) and pH strips. Record results.
5. Compare: do “hazardous”-flagged devices show measurably higher contamination (lead/copper presence, pH shift) than “safe”-flagged devices? Run this across enough samples (aim for at least 3 devices per hazard category, ideally more) to support a real comparison, not just a couple of anecdotal readings.
6. This step directly tests whether your engineering tool’s hazard label predicts something real, rather than being an assumption.

Checkpoint: A dataset comparing leachate contamination levels between hazard-flagged and safe-flagged devices — your environmental science validation result.

Phase 5 — Real-World Validation of the Engineering Tool (Weeks

8-9)

1. Once your validation partner (arranged in Step 4 of “this week”) is ready, run your full non-destructive protocol on their real donated e-waste batch — without opening any devices.
2. Use your trained model to predict value and hazard flags for this new, external batch.
3. If allowed, select a small subset (5-10 items) for actual teardown to check real-world model accuracy on unseen data.

Checkpoint: Evidence your tool generalizes beyond your own curated sample — the step that elevates this from a home project to a field-tested one.

Phase 6 — Analysis & Writeup (Weeks 9-11)

1. Report results in three clearly separated sections: (a) engineering model performance on your teardown-validated dataset, (b) leaching/contamination validation results (does the hazard label predict real contamination?), (c) model performance on the real external batch.
 2. Be explicit about limitations: small sample sizes at each stage, test-strip precision vs. lab-grade assay, and how confidence would scale with more data/replicates.
 3. Frame your narrative clearly: a real gap (no accessible non-destructive triage method exists) → your novel low-cost engineering solution → a genuine scientific test of whether its hazard predictions reflect real environmental risk → a real-world field test proving it’s not just theoretical.
-

Buffer (Weeks 11+)

Reserve 1-2 weeks for board/paper polish and mock judging. Prepare answers for: “How did you choose your ground-truth value estimates?”, “How sensitive are your heavy-metal test strips compared to lab assay, and how does that affect your conclusions?”, and “What would you add with more resources?”

Materials & Budget

Item	Cost
Digital multimeter	\$15-20
Magnet / Hall-effect sensor module	\$5-10
Precision scale	\$15-20
Screwdriver set (if not owned)	\$10-15
Heavy metal (lead/copper) test strips	\$15-25
pH test strips	\$5
Clear soaking containers	\$10
Misc. (labels, spreadsheet/logging supplies)	\$5-10
Total	~\$80-120

No bulk chemical handling — test strips are consumer-safe, and the only “hazard” exposure is incidental to devices you’re already opening carefully.

Updated Timeline Summary

Weeks	Phase
1-2	Non-destructive sensing protocol on validation sample
2-4	Manual teardown ground truth
4-6	Build & validate prediction model
6-8	Leaching validation (environmental science layer)
8-9	Real-world field validation
9-11	Analysis and writeup

This Week's Checklist

- ☐ Order multimeter, magnet/Hall sensor, precision scale
- ☐ Order heavy metal test strips + pH strips + soaking containers
- ☐ Start collecting 15-25 devices for your validation sample
- ☐ Reach out to a school/library/community center/e-waste drive about real-world validation access (start now)
- ☐ Install Python + scikit-learn (+ PyTorch if using image input)
- ☐ Set up your spreadsheet template for logging device readings